

FORMALISING VON NEUMANN'S PROOF OF THE RADON-NIKODYM THEOREM

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INTRODUCTION

Given a measurable space $(\Omega, \mathcal{F})$, and measures ν and μ such that ν is absolutely continuous with respect to μ , i.e. for all $A \in \mathcal{F}$:

$$\mu\{A\} = 0 \implies \nu\{A\} = 0, \quad \text{denoted } \nu \ll \mu,$$

then the Radon-Nikodym theorem asserts that there exists a $\mathcal{F}$ -measurable function, called the Radon-Nikodym derivative and denoted by

$$\frac{d\nu}{d\mu} : \Omega \longrightarrow [0, \infty),$$

s.t. ν can be expressed as an integral of such function with respect to the measure μ , that is, for any $A \in \mathcal{F}$:

$$\nu\{A\} = \int_A \frac{d\nu}{d\mu} d\mu.$$

We can interpret this result as a “representation theorem”: in fact, von Neumann re-derived it from a central representation result of his theory of Hilbert spaces. Recall that a Hilbert space $\mathbf{H}$ is just a reflexive Banach space endowed with an inner product $(\cdot, \cdot)$ generating its norm:

$$\|f\|^2 = (f, f), \quad x \in \mathbf{H}$$

and that on a Banach space $\mathbf{B}$, i.e. a complete normed vector space, one usually considers the action of linear functionals $L \in \mathbf{B}^*$ as the following duality pairing:

$$\langle L, f \rangle \in \mathbb{R}, \quad f \in \mathbf{B}$$

while the Banach space is said to be reflexive when there is an isomorphism $\mathbf{B} \simeq (\mathbf{B}^*)^*$ and, in particular:

$$\langle L^*, L \rangle = \langle L, f \rangle$$

for any $L \in \mathbf{B}$ and $L^* \in (\mathbf{B}^*)^*$.

FRÉCHET-RIESZ REPRESENTATION THEOREM. Given any bounded linear functional $L \in \mathbf{H}^*$, then there exists a unique representative $\xi \in \mathbf{H}$ such that, for any $f \in \mathbf{H}$:

$$\langle L, f \rangle = (\xi, f).$$

The full argument of von Neumann can be found in the monograph of Peter D. Lax on Functional Analysis [Lax14], and it goes as follows: the Hilbert space is $\mathbf{L}^2(\pi)$, that of square integrable functions with respect to the measure $\pi = \mu + \nu$. In Lean4:

```
noncomputable def π : Measure Ω := μ + ν
local notation "L2(π)" => Lp ℝ 2 (π μ ν)
```

that is, we are looking at the $L^p(\Omega; \mathbb{R})$ space consisting of equivalence classes (quotient space under the equivalence relation of being a.e. identically equal) of measurable functions where the p^{th} power of the absolute value is π -integrable, with the choice of $p = 2$. We are namely considering the inner product

$$(f, g) = \int_{\Omega} f g \, d\pi$$

and thus the norm

$$\|f\| = \int_{\Omega} f^2 \, d\pi.$$

In Mathlib4, this is defined as follows:

```
def eLpNorm' {α : MeasurableSpace Ω} (f : Ω → α) (p : ℝ) (μ : Measure Ω) : ℝ≥0∞ :=
  (∫- x, ‖f x‖e ^ p dμ) ^ (1 / p)
```

where $\int^-$ represents the lower Lebesgue integral of the function and $\|\cdot\|_e$ the extended non-negative real number given by the modulus of an extended real number.

I could have just worked by taking a function f and declaring it to be square-integrable with respect to the measure π , whereas I decided to let f be a fixed element of the ring of functions $\mathbf{L}^2(\pi)$: at first, I started my project with the first approach but, soon enough, I went and checked what I would have needed to use the Fréchet-Riesz representation theorem, which corresponds to the `InnerProductSpace.toDual` result in Mathlib4 and requires an inner product space. This will be applied to the bounded linear functional

$$\langle L, f \rangle = \int_{\Omega} f \, d\nu, \quad f \in \mathbf{L}^2(\pi)$$

yielding a representative $\xi \in \mathbf{L}^2(\pi)$ s.t.

$$\int_{\Omega} f \, d\nu = \langle L, f \rangle = (\xi, f) = \int_{\Omega} \xi f \, d\pi. \quad (*)$$

This in particular implies the Radon-Nikodym theorem by letting

$$\frac{d\nu}{d\mu} = \frac{\xi}{1 - \xi}$$

as we can rewrite (*) as follows:

$$\int_{\Omega} f \, d\nu = \int_{\Omega} \xi f \, d\mu + \int_{\Omega} \xi f \, d\nu \quad \text{i.e.} \quad \int_{\Omega} (1 - \xi) f \, d\nu = \int_{\Omega} \xi f \, d\mu$$

and it is enough to choose $\xi f \equiv \chi_A$ for $A \in \mathcal{F}$. We in fact then get, multiplying on both sides by $\xi/(1 - \xi)$:

$$\nu\{A\} = \int_{\Omega} \chi_A \, d\nu = \int_{\Omega} \chi_A \frac{d\nu}{d\mu} \, d\mu = \int_A \frac{d\nu}{d\mu} \, d\mu$$

which is the desired claim. The crucial assumption that $\nu \ll \mu$ seems to be unused, but the argument would be invalid without that, as it ensures our Radon-Nikodym representative to be well-defined: in particular we can modify, if necessary, the function ξ on a set of μ -measure zero so that $0 \leq \xi < 1$ holds everywhere* and, by absolute continuity, this does not affect (*).

Our aim is here to provide a formalisation of the proof of the fact that $L : \mathbf{L}^2(\pi) \rightarrow \mathbb{R}$ is a bounded linear functional, namely, that it satisfies the assumption of the Fréchet-Riesz representation theorem, so that the Radon-Nikodym derivative (in our case, the representative ξ will be enough) can be extracted.

MEASURE-THEORETIC PRELIMINARY RESULTS

At the beginning, I thought my project was going to be short, being composed of the two results that L is linear and bounded/continuous. However these, especially the second, required some background work which we collect here. I found the couple of relevant Mathlib4 known results in Measure Theory, as well as some needed to make computations with ENNReals, with the aid of LeanDex.

First, I proved that $\nu \leq \pi = \mu + \nu$ as measures, i.e. “pointwise” on each measurable set, and that $\nu \ll \pi$; these two lemmas respectively correspond to the `le_add_left` and `absolutelyContinuous_of_le` results in Mathlib’s sections on Measures. The second clearly relies on the first one as, if $\nu \leq \pi$ then, for all $A \in \mathcal{F}$:

$$\pi\{A\} = 0 \geq \nu\{A\}$$

implies $\nu\{A\} = 0$, i.e. $\nu \ll \pi$. From the first, it also follows that our fixed $f \in \mathbf{L}^2(\pi)$ is ν -square integrable too, via the following lemma:

```
MeasureTheory.MemLp.mono_measure {π ν : Measure Ω} {f : Ω → ℝ}
  (hπν : ν ≤ π) (hf : MemLp f p π) : MemLp f p ν
```

and, though trivial for an informal proof, the fact that the $\mathbf{L}^2(\nu)$ norm is bounded by the $\mathbf{L}^2(\pi)$ turned out to be much harder to formalise. This is due to the fact that I would have needed this result to later prove the boundedness of the functional, which, as discussed below, I defined using Lean’s Bochner integral, namely I found myself with the following goal:

It straightforwardly holds μ -a.e. as, letting A, B be respectively the sets on which $\xi < 0$ and $\xi \geq 1$, by () with $f \equiv \chi_A, \chi_B$, one gets that A, B are sets of μ -measure zero.

```
lemma norm_le_of_measure_le (f : L2(π)) :
  (∫ x, ‖f x‖ ^ (2 : ℝ) ∂v) ^ (1/2 : ℝ) ≤ ‖f‖
```

where, unravelling the definition of the $\mathbf{L}^2(\pi)$ norm $\|f\|$, that is actually expressed as a lower Lebesgue integral. So, I had to struggle a bit to match the two[†]. Squaring both sides and suitably rewriting and cleaning the integral definition of the $\mathbf{L}^2(\pi)$ norm $\|f\|$ on the RHS, our goal looks like:

```
∫ (x : Ω), ‖↑↑f x‖ ^ 2 ∂v ≤ ‖f‖ ^ 2 = (∫ (x : Ω), ‖↑↑f x‖_e ^ 2 ∂π μ v).toReal
```

and we face a nasty double coercion on the function $f \in \mathbf{L}^2(\pi)$. I was very confused by this at first, as I expected one coercion only: the outer one, from a space of equivalence classes of a.e. strongly measurable functions to functions. What Lean is doing is extracting one representative out of the equivalence class $f \in \mathbf{L}^2(\pi)$ considered, the type changing from $\Omega \rightarrow_m [\pi \mu v] \mathbb{R}$ to $\Omega \rightarrow \mathbb{R}$. That will not affect our goal because the integral of each element in that equivalence class is π -a.e. (and thus v -a.e.) identical and their π -integral (and thus v -integral) the same. What I did not realise, and I understood by asking questions during Office Hours and Labs, is that in Lean an element $f \in \mathbf{L}^2(\pi)$ is not only an equivalence class, as it comes bundled with the mathematical proof that the function is square-integrable. The inner coercion drops the proof and leaves solely the equivalence class, the type changing from $\uparrow \mathbf{L}^2(\pi)$ to $\Omega \rightarrow_m [\pi \mu v] \mathbb{R}$.

Now, we can use the following lemma:

```
MeasureTheory.integral_eq_lintegral_of_nonneg_ae {v : Measure Ω} {f : Ω → ℝ}
(hf : 0 ≤[ae v] f) (hfm : AESTronglyMeasurable f v) :
  ∫ (x : Ω), f x ∂v = (∫ (x : Ω), ENNReal.ofReal (f x) ∂v).toReal
```

to rewrite the LHS as a lower Lebesgue integral. The second needed assumption `(right.hfm)` that f^2 is v -strongly measurable is easy to check:

```
/- f is π-strongly measurable by definition -/
have h_meas_pi := Lp.aestronglyMeasurable f
/- f is v-strongly measurable because v ≤ π -/
have h_meas_nu : AESTronglyMeasurable f v :=
  h_meas_pi.mono_measure (monotonicity_measure μ v)
exact h_meas_nu.pow 2 -- then f^2 is v-strongly measurable □
```

[†]The LHS' Bochner integral coincides with the corresponding lower Lebesgue integral because we are looking at real-valued (rather than general Banach-valued), non-negative integrand functions.

and it was in fact suggested me in a Lab that I could just use the `fun_prop` tactic for such trivial statements about measurability of functions. I was similarly suggested the `filter_upwards` tactic to prove that the second assumption `(right.hf)` that $f^2 \geq 0$ ν -a.e. actually holds pointwise. For the reduced goal:

$$\int_{\Omega} (x : \mathbb{Q}), \text{ENNReal.ofReal } (\uparrow \uparrow f \ x \ ^2) \ d\nu \leq \int_{\Omega} (x : \mathbb{Q}), \|\uparrow \uparrow f \ x\|_e^2 \ d\pi \ \mu \ \nu$$

we can use the `lintegral_mono'` result of monotonicity of the lintegral with respect to the measure. We are left with the residual check that $\int_{\Omega} \|f\|_e^2 \ d\pi$ is finite, via the following:

$$\text{ENNReal.toReal_mono } \{a \ b : \text{ENNReal}\} \ (hb : b \neq \top) \ (h : a \leq b) : a.\text{toReal} \leq b.\text{toReal}$$

but that is easy because the $\mathbf{L}^2(\pi)$ norm of f is finite by definition.

THE BOUNDED LINEAR FUNCTIONAL

We define the von Neumann functional, as follows:

```
noncomputable def functional (f : L2( $\pi$ )) :  $\mathbb{R}$  :=  $\int x, f \ x \ d\nu$ 
local notation "L" => functional  $\mu \ \nu$ 
```

so that the integral involved in the aforementioned definition is formalised as a Bochner integral. This is an obliged choice because the integrand is not necessarily (ν -a.e.) non-negative and so the lower Lebesgue integral may not be well-defined. This is the source of the pain given to us in the preliminary results. We moreover assume ν is a finite measure, i.e. $\nu\{\Omega\}$ is finite and, in particular, $\nu\{A\}$ is finite for all measurable sets $A \subseteq \Omega$, as it is crucial for the von Neumann functional L to be both linear and bounded, which we now prove.

Regarding the linearity, the homogeneity, i.e. the goal:

$$\forall (c : \mathbb{R}) (f : \uparrow L^2(\pi)), L(c \cdot f) = c \cdot L f$$

is easier. The notation $\cdot$ is type-dependent: for the RHS, that formalises the product between real numbers via the following Mathlib lemma:

$$\text{smul_eq_mul}.\{u_9\} \ \{\alpha : \text{Type } u_9\} \ [\text{Mul } \alpha] \ (a \ b : \alpha) : a \cdot b = a * b$$

which is saying that, given elements of a type equipped with some multiplication operation, such as $(\mathbb{R}, \cdot)$, then the type-dependent product $\cdot$ corresponds to such operation. For the LHS instead, we need to prove the following:

$$\int x, (c \cdot f) \ x \ d\nu = \int x, c * f \ x \ d\nu$$

as that would allow us to bring out the multiplicative constant with `integral_const_mul`. Here we use absolute continuity, indeed, thanks to the fact that $\nu \ll \pi$, via the Mathlib lemma:

```
MeasureTheory.Measure.AbsolutelyContinuous.ae_eq {π ν : Measure Ω} (h : ν << π)
{f g : Ω → ℝ} (h' : f =[ae π] g) : f =[ae ν] g
```

it is enough to show that

```
↑↑(c • f) =[ae π] c • ↑↑f
```

but this is exactly the `Lp.coeFn_smul` result, which I would call “homogeneity of the double coercion[‡] with respect to the type-dependent product”, and we are done. For the additivity part, i.e. the goal:

```
∀ (f g : ↑L2(π)), L (f + g) = L f + L g
```

we can similarly rewrite the LHS similarly to what we did for the homogeneity, i.e. by absolutely continuity we reduce to proving:

```
∫ (x : Ω), (f + g) x ∂ν = ∫ (x : Ω), f x + g x ∂ν
```

by means of the `Lp.coeFn_add` result, which we can in turn call “homogeneity of the double coercion with respect to addition”. But then, we are done thanks to additivity `MeasureTheory.integral_add` of the Bochner integral: that requires f, g to be both ν -integrable, which is true because they are ν -strongly measurable (now, I just went for an easy `fun_prop` to check that) and have a finite ν -integral.

The remaining hypothesis of the Fréchet-Riesz representation theorem is the boundedness, or continuity, of the von Neumann functional, i.e.

$$\exists C > 0 \quad \text{s.t.} \quad |\langle L, f \rangle| \leq C \|f\|$$

which is a straightforward computation

$$\begin{aligned} |\langle L, f \rangle| &\leq \int_{\Omega} |f| \, d\nu = \int_{\Omega} |f| \cdot 1 \, d\nu = \int_{\Omega} |f| \cdot |1| \, d\nu \\ (\text{Hölder}) \quad &\leq \left(\int_{\Omega} |f|^2 \, d\nu \right)^{1/2} \cdot \left(\int_{\Omega} |1|^2 \, d\nu \right)^{1/2} \\ (\nu \leq \pi) \quad &\leq \left(\int_{\Omega} |f|^2 \, d\pi \right)^{1/2} \cdot \nu\{\Omega\}^{1/2} = \sqrt{\nu\{\Omega\}} \cdot \|f\| \end{aligned}$$

so it is enough to choose $C = \sqrt{\nu\{\Omega\}} + 1$. We formalise this using the `calc` tactic. In the crucial fourth step, we apply Lean’s version `MeasureTheory.integral_mul_norm_le_Lp_mul_Lq` of Hölder’s inequality for the Bochner integral.

[‡]This specific double coercion from the pairs of equivalence class and integrability proof, to equivalence classes, to functions.

To apply Lean’s version `InnerProductSpace.toDualMap` of the Fréchet-Riesz representation theorem, we bundle the von Neumann functional into a continuous real-valued linear map, of type $L^2(\pi) \rightarrow L[\mathbb{R}] \mathbb{R}$ that is bounded in the sense proved:

```

noncomputable def L_cont : L2(π) → L[ℝ] ℝ :=
  let boundedness := L_bounded μ v
  let C := Classical.choose boundedness, hC := Classical.choose_spec boundedness
  LinearMap.mkContinuous
    { toFun := fun f ↦ L f
      map_add' := (L_linear μ v).map_add
      map_smul' := (L_linear μ v).map_smul }
  C, hC.2

```

where `LinearMap.mkContinuous` effectively constructs such a bounded linear functional. Then it admits a representation as an inner product:

```

noncomputable def ξ : L2(π) :=
  (InnerProductSpace.toDual ℝ L2(π)).symm (L_cont μ v)

```

which was our objective. The `InnerProductSpace.toDual` result is exactly assigning, to a given vector in the given Hilbert space, the functional given by $f \mapsto \langle f, v \rangle$. This is an isometry, which we reverse using `symm` and apply to our bounded linear functional.

CONCLUSIONS

While feeling a bit overwhelmed at the beginning by the complicated goals I kept getting with different integrals and `ENNReals`, I slowly came to grasp the logic of how these things are defined and work inside `Mathlib`. Also, I was very satisfied that I could continue with the proofs by just using `LeanDex` heavily to find names of useful lemmas in `Mathlib`. When I could not, I went to Office Hours and Labs and got help there, especially with the proof of the linearity: there, I was guided on how to prove additivity and later at home I proved homogeneity by myself. I was also very happy of learning new tactics such as `constructor`, `filter upwards` and `fun_prop`. It was suggested to me, in a Lab, that I had to bundle the functional into the bounded linear functional `L_cont` of type $L^2(\pi) \rightarrow L[\mathbb{R}] \mathbb{R}$, and, to actually prove the representation identity: $\langle L, f \rangle = (\xi, f)$, I was struggling a bit with writing the inner product and I got huge help by Google Gemini, in this conversation.

REFERENCES

[Lax14] Peter D. Lax. *Functional Analysis*. John Wiley & Sons, 2014.